Science

Nature Notebook
Nature Walks & Scouting
Introduction to Engineering Lessons
Introduction to Engineering Labs

Alveary

Alveary

Alt. Science Options

Science: Introduction to Engineering

GRADE(S): 9-12

About the Course

This course includes the following topic(s): Introduction to Engineering Lessons, Introduction to Engineering Labs, Nature Notebook: Grades 9-12, Nature Walks & Scouting: Grades 9-12

About Nature Notebook: Grades 9-12
Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

About Nature Walks & Scouting: Grades 9-12
Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

About Introduction to Engineering Lessons
An elective in Physical Science, Introduction to Engineering develops thinking skills and practical experience that are applicable to any field of design/innovation, while incorporating historical context, modern developments, current events, and citizenship. This is a VERY hands-on course that requires independent interest in any form of design/innovation, including various fields of engineering and the skilled trades. The course provides guidance and flexibility for teachers and learners to adjust the course for personal needs and preferences.

About Introduction to Engineering Labs
Note that labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.

Placement & Combining Tips

Nature Notebook: Grades 9-12
Learners may be combined and follow their own interests.

Nature Walks & Scouting: Grades 9-12
Learners may follow their own interests or follow the plan of their local scouting troop or natural history club.

Introduction to Engineering Lessons
The completion of Algebra 1 and Geometry are recommended for this course, as the experience will contribute to the learner's understanding, but they are not required.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
9-12	Nature Notebook: Grades 9-12 1+ time/week 20 min+	
9-12	Nature Walks & Scouting: Grades 9-12 1 time/week 30 min+	
9-12	Introduction to Engineering Lessons 5 times/week 45 min	The Things We Make The Boy Who Harnessed the Wind The Boy Who Harnessed the Wind, Young Reader's Edition

GRADE	SCHEDULE INFO.	BOOKS
		The Way Things Work: Newly Revised Edition: The Ultimate Guide to How Things Work By Design: Ethics, Theology, and the Practice of Engineering
9-12	Introduction to Engineering Labs 1 time/week 60 min	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Introduction to Engineering				
Introduction to Engineering Lessons Nature Walks & Scouting: Grades 9-12	Introduction to Engineering Lessons	Introduction to Engineering Lessons	Introduction to Engineering Lessons Nature Notebook: Grades 9-12	Introduction to Engineering Lessons Introduction to Engineering Labs

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Introduction to Engineering Lessons
☐ There are 2 sensitive terms used in Ch.2 of The Things We Make as part of a larger discussion on design bias. Teachers should review Lesson 1 before the start of the term.

☐ Carve out time to continue or establish the regular habit of spending time in nature, including the use of a nature journal, as appropriate.

☐ Obtain materials from the supply lists.

☐ Select a science book from the Alveary bookshelf for personal reading time, as appropriate.

☐ Bookmark or print Quick Links, as needed.

Term Prep & Teacher Tips

Introduction to Engineering Lessons
☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
• items with which to tinker by student choice; Basements, closets, thrift shops, and scrap exchanges are great places to find objects for tinkering
• materials for projects by student choice; if you would like to have materials for the suggested Week 1 activity, obtain the optional supplies below
• optional: 2-3 boxes uncooked spaghetti (Term 1 Week 1)
• optional: all-purpose glue (Term 1 Week 1)
• optional: uncooked spaghetti (Term 1 Week 1)
• optional: S hook (Term 1 Week 1)
• optional: plastic bucket (Term 1 Week 1)
• optional: water or sand (Term 1 Week 1)
• optional: uncooked spaghetti (Term 1 Week 1)
• optional: kitchen scale(Term 1 Week 1)
• optional: bathroom scale (Term 1 Week 1)
• optional: 6-12" piece of wooden board (Term 1 Week 1)

Books & Resources

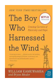
For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ View Book List Details

Science: Introduction to Engineering

Introduction to Engineering Lessons


The Things We Make


The Boy Who Harnessed the Wind


The Boy Who Harnessed the Wind, Young Reader's Edition


The Way Things Work: Newly Revised Edition: The Ultimate Guide to How Things Work


By Design: Ethics, Theology, and the Practice of Engineering

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ View Basic Supplies

(No Subject Supplies Assigned)

Quick Links

Click THIS text or scan the QR code for links.

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Science: Introduction to Engineering

Introduction to Engineering Lessons

∞ Science News Explores

Science: Introduction to Engineering

How To Approach

Introduce

• If starting a new book or a new topic in the book, then look at the title or a picture and take a moment to consider previous ideas and experiences.

• If continuing a previous reading, recap what was read previously. Often, the title of the book's section can help to draw out the main idea.

Read

• Read or do, as instructed in the lessons, making note of or flagging unfamiliar terms, interesting ideas, important dates, inspiring quotes, etc.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• As they read, learners record ideas in a notebook or binder using outlines, diagrams, graphic organizers, or other methods (or a combination of methods) that suit them. These recordings can be a helpful mechanism for remembering or a mini-narration to support understanding.

• If learners do not understand a word or concept, do not worry. Try reading over the passage again, studying a picture or diagram, connecting the idea to something from real life, or practicing chapter exercises. The lab/field work further supports major concepts from the text.

Narrate

• Process the ideas of the lesson by retelling, defining a concept, explaining the links in a chain of thought, etc. Do this orally or silently to yourself.

• Use words, pictures, outlines, etc.

• If a particular idea cannot be narrated, then read or examine the text again.

Discuss

• Consider with the teacher any thoughts, confusion, or concerns about the passage.

• If understanding is still uncertain, try rereading the passage or do some personal research on the topic.

Connect

• Follow any extra links, examine any sidebars in the text, or pursue additional reading, depending on student interest.

Science: Introduction to Engineering

How To Approach

Introduce

• Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.

• Relevant concept(s) are introduced in the text.

• Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.

Lab Procedure

• Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.

• Record all data and observations in the lab notebook.

Analysis & Conclusions

• After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.

• If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

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Term 1

WEEK 1

⬚ **45m Introduction to Engineering Lessons - Lesson 1**
How Engineering Works

Materials: The Things We Make

PREP: Read Teacher Tip

→ INTRO
From medieval cathedrals to modern soda cans, this book will help you grasp "the engineering method" and notice it working all around you. Note that during a discussion regarding bias in the design process (Ch.2), the author uses two terms that may be unfamiliar to you:
porn - short for pornography; inappropriate, sexually-oriented media that treats people as material objects rather than whole persons.
cisgender - a modern term used to describe people whose biological sex and sociological gender are aligned, as opposed to transgender.

→ READ, NARRATE, & DISCUSS
Introduction

→ PREPARE
Since an important aspect of engineering is tinkering and developing your intuition, spend some time looking for an item you'd like to work with on your Tinker Lab days. This could be a broken clock, coffee maker, soap dispenser, etc. Anything found in a closet or a thrift store that you can take apart and figure out without worrying your teacher is the perfect thing!

★ TEACHER TIP
There are two sensitive terms used in Ch.2, which some students may not be familiar with. The author only mentions them in a larger discussion about the impact of design bias. Nothing beyond the basic definitions provided in today's introduction is needed, but you may want to preview pp.30, 42 if you would like to understand the context or consider any additional discussion you would like to introduce.

• PLAN WEEKLY
⬚ take a nature walk
⬚ science free read

WEEK 1

⬚ **45m Introduction to Engineering Lessons - Lesson 2**
Tinker Lab

Materials: by student choice & notebook

→ INTRO
You have two days each week to tinker with found objects that you have permission to take apart, figure out, and attempt to put back together (or improve). Basements, closets, thrift shops, and scrap exchanges are great places to find objects for tinkering. When you finish with one object, simply find another!

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 1

⬚ **45m Introduction to Engineering Lessons - Lesson 3**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 1

⬚ **45m Introduction to Engineering Lessons - Lesson 4**
Engineering Literature

Materials: The Boy Who Harnessed the Wind

→ READ, NARRATE, & DISCUSS
about 20 pages or 1 chapter per week

WEEK 1

⬚ **45m Introduction to Engineering Lessons - Lesson 5**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event (link below and in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?' A fun TEDEd video is provided for inspiration.
∞ Video Link: TEDEd (6:19)
∞ Website Link: Science News Explores

WEEK 1

⬚ **60m Introduction to Engineering Labs - Lesson 1**

Materials: internet access

→ PROJECT
If there is a design problem in your home or community that you would like to tackle, use this time to research, design, and execute your solution.

If you'd like to sample different areas of engineering, begin with the structural engineering project

Science: Introduction to Engineering

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Term 1

linked, The Leaning Tower of Pasta. Or you might find your own idea by exploring the Science Buddies website. The filters on the left side of the page allow you to narrow your results. Once you have chosen a project, let your teacher know if you need any materials. Your project might take the whole year or you might finish it in just a few weeks. Whenever you finish your project, select another one! Record your work and your observations in your lab notebook.
∞ Website Link: The Leaning Tower of Pasta
∞ Website Link: Science Buddies

WEEK 2

⬚ **45m Introduction to Engineering Lessons - Lesson 6**
How Engineering Works

Materials: The Things We Make

→ READ, NARRATE, & DISCUSS
Ch.1

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 2

⬚ **45m Introduction to Engineering Lessons - Lesson 7**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 2

⬚ **45m Introduction to Engineering Lessons - Lesson 8**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 2

⬚ **45m Introduction to Engineering Lessons - Lesson 9**
Engineering Literature

Materials: The Boy Who Harnessed the Wind

→ READ, NARRATE, & DISCUSS
about 20 pages or 1 chapter per week

WEEK 2

⬚ **45m Introduction to Engineering Lessons - Lesson 10**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 2

⬚ **60m Introduction to Engineering Labs - Lesson 2**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 3

⬚ **45m Introduction to Engineering Lessons - Lesson 11**
How Engineering Works

Materials: The Things We Make

→ READ, NARRATE, & DISCUSS
Ch.2

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 3

⬚ **45m Introduction to Engineering Lessons - Lesson 12**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

Science: Introduction to Engineering

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QR

Term 1

WEEK 3

⬚ **45m Introduction to Engineering Lessons - Lesson 13**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 3

⬚ **45m Introduction to Engineering Lessons - Lesson 14**
Engineering Literature

Materials: The Boy Who Harnessed the Wind

→ READ, NARRATE, & DISCUSS
about 20 pages or 1 chapter per week

WEEK 3

⬚ **45m Introduction to Engineering Lessons - Lesson 15**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 3

⬚ **60m Introduction to Engineering Labs - Lesson 3**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 4

⬚ **45m Introduction to Engineering Lessons - Lesson 16**
How Engineering Works

Materials: The Things We Make

→ READ, NARRATE, & DISCUSS
Ch.3

→ SUPPLEMENTAL
∞ Article Link: The Story of Radia Pearlman's Spanning Tree

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 4

⬚ **45m Introduction to Engineering Lessons - Lesson 17**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 4

⬚ **45m Introduction to Engineering Lessons - Lesson 18**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 4

⬚ **45m Introduction to Engineering Lessons - Lesson 19**
Engineering Literature

Materials: The Boy Who Harnessed the Wind

→ READ, NARRATE, & DISCUSS
about 20 pages or 1 chapter per week

WEEK 4

⬚ **45m Introduction to Engineering Lessons - Lesson 20**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might

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Term 1

read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 4

☐ **60m Introduction to Engineering Labs - Lesson 4**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project

WEEK 5

☐ **45m Introduction to Engineering Lessons - Lesson 21**
How Engineering Works

Materials: The Things We Make

→ READ, NARRATE, & DISCUSS
Ch.4

- **PLAN WEEKLY**
  ☐ take a nature walk
  ☐ science free read

WEEK 5

☐ **45m Introduction to Engineering Lessons - Lesson 22**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 5

☐ **45m Introduction to Engineering Lessons - Lesson 23**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 5

☐ **45m Introduction to Engineering Lessons - Lesson 24**
Engineering Literature

Materials: The Boy Who Harnessed the Wind

→ READ, NARRATE, & DISCUSS
about 20 pages or 1 chapter per week

WEEK 5

☐ **45m Introduction to Engineering Lessons - Lesson 25**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 5

☐ **60m Introduction to Engineering Labs - Lesson 5**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 6

☐ **45m Introduction to Engineering Lessons - Lesson 26**
How Engineering Works

Materials: The Things We Make

→ READ, NARRATE, & DISCUSS
Ch.5

- **PLAN WEEKLY**
  ☐ take a nature walk
  ☐ science free read

WEEK 6

☐ **45m Introduction to Engineering Lessons - Lesson 27**
Tinker Lab

Materials: by student choice & notebook

→ TINKER

Science: Introduction to Engineering

Click THIS text or scan the QR code for links.

QR

Term 1

Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 6

⬚ **45m Introduction to Engineering Lessons - Lesson 28**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 6

⬚ **45m Introduction to Engineering Lessons - Lesson 29**
Engineering Literature

Materials: The Boy Who Harnessed the Wind

→ READ, NARRATE, & DISCUSS
about 20 pages or 1 chapter per week

WEEK 6

⬚ **45m Introduction to Engineering Lessons - Lesson 30**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 6

⬚ **60m Introduction to Engineering Labs - Lesson 6**

Materials: by student choice & notebook

→ PROJECT
If there is a design problem in your home or community that you would like to tackle, use this time to research, design, and execute your solution.

If you'd like to sample different areas of engineering, try the materials engineering project linked, Pounding Papyrus. Or you might find your own idea by exploring the Science Buddies website. The filters on the left side of the page allow you to narrow your results. Once you have chosen a project, let your teacher know if you need any materials. Your project might take the whole year or you might finish it in just a few weeks. Whenever you finish your project, select another one! Record your work and your observations in your lab notebook.
∞ Website Link: Pounding Papyrus
∞ Website Link: Science Buddies

WEEK 7

⬚ **45m Introduction to Engineering Lessons - Lesson 31**
How Engineering Works

Materials: The Things We Make

→ READ, NARRATE, & DISCUSS
Ch.6

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 7

⬚ **45m Introduction to Engineering Lessons - Lesson 32**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 7

⬚ **45m Introduction to Engineering Lessons - Lesson 33**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 7

⬚ **45m Introduction to Engineering Lessons - Lesson 34**
Engineering Literature

Materials: The Boy Who Harnessed the Wind

→ READ, NARRATE, & DISCUSS

Science: Introduction to Engineering

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Term 1

about 20 pages or 1 chapter per week

WEEK 7

⬚ **45m Introduction to Engineering Lessons - Lesson 35**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ **NOTE**
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 7

⬚ **60m Introduction to Engineering Labs - Lesson 7**

Materials: by student choice & notebook

→ **PROJECT**
Continue working on your project.

WEEK 8

⬚ **45m Introduction to Engineering Lessons - Lesson 36**
How Engineering Works

Materials: The Things We Make

→ **READ, NARRATE, & DISCUSS**
Ch.7

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 8

⬚ **45m Introduction to Engineering Lessons - Lesson 37**
Tinker Lab

Materials: by student choice & notebook

→ **TINKER**
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 8

⬚ **45m Introduction to Engineering Lessons - Lesson 38**
Tinker Lab

Materials: by student choice & notebook

→ **TINKER**
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 8

⬚ **45m Introduction to Engineering Lessons - Lesson 39**
Engineering Literature

Materials: The Boy Who Harnessed the Wind

→ **READ, NARRATE, & DISCUSS**
about 20 pages or 1 chapter per week

WEEK 8

⬚ **45m Introduction to Engineering Lessons - Lesson 40**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ **NOTE**
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 8

⬚ **60m Introduction to Engineering Labs - Lesson 8**

Materials: by student choice & notebook

→ **PROJECT**
Continue working on your project.

Science: Introduction to Engineering

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Term 1

WEEK 9

☐ **45m Introduction to Engineering Lessons - Lesson 41**
How Engineering Works

Materials: The Things We Make

→ READ, NARRATE, & DISCUSS
Ch.8

- **PLAN WEEKLY**
- ☐ take a nature walk
- ☐ science free read

WEEK 9

☐ **45m Introduction to Engineering Lessons - Lesson 42**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 9

☐ **45m Introduction to Engineering Lessons - Lesson 43**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 9

☐ **45m Introduction to Engineering Lessons - Lesson 44**
Engineering Literature

Materials: The Boy Who Harnessed the Wind

→ READ, NARRATE, & DISCUSS
about 20 pages or 1 chapter per week

WEEK 9

☐ **45m Introduction to Engineering Lessons - Lesson 45**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 9

☐ **60m Introduction to Engineering Labs - Lesson 9**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 10

☐ **45m Introduction to Engineering Lessons - Lesson 46**
How Engineering Works

Materials: The Things We Make

→ READ, NARRATE, & DISCUSS
Ch.9

- **PLAN WEEKLY**
- ☐ take a nature walk
- ☐ science free read

WEEK 10

☐ **45m Introduction to Engineering Lessons - Lesson 47**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 10

☐ **45m Introduction to Engineering Lessons - Lesson 48**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

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Term 1

WEEK 10	⬚ **45m Introduction to Engineering Lessons - Lesson 49** *Engineering Literature* Materials: The Boy Who Harnessed the Wind **→ READ, NARRATE, & DISCUSS** about 20 pages or 1 chapter per week	
WEEK 10	⬚ **45m Introduction to Engineering Lessons - Lesson 50** *Present What You've Learned* **→ PREPARE** For one of your exam questions, you will be asked to give an informal presentation to your teacher on something you have learned about this term. This could be how a particular object is made or constructed, or something else you have researched. Use this time to gather your thoughts, think through how you want to do that, and work on it. You can be as creative as you would like.	
WEEK 10	⬚ **60m Introduction to Engineering Labs - Lesson 10** Materials: by student choice & notebook **→ PROJECT** Continue working on your project.	
WEEK 11	⬚ **45m Introduction to Engineering Lessons - Lesson 51** *How Engineering Works* Materials: The Things We Make **→ READ, NARRATE, & DISCUSS** Read the Afterward section.	• **PLAN WEEKLY** ⬚ take a nature walk ⬚ science free read
WEEK 11	⬚ **45m Introduction to Engineering Lessons - Lesson 52** *Tinker Lab* Materials: by student choice & notebook **→ TINKER** Take apart, figure out, and attempt to put back together (or improve). Record your observations.	
WEEK 11	⬚ **45m Introduction to Engineering Lessons - Lesson 53** *Tinker Lab* Materials: by student choice & notebook **→ TINKER** Take apart, figure out, and attempt to put back together (or improve). Record your observations.	
WEEK 11	⬚ **45m Introduction to Engineering Lessons - Lesson 54** *Engineering Literature* Materials: The Boy Who Harnessed the Wind **→ NOTE** We'll finish this book in week 4 of next term! **→ READ, NARRATE, & DISCUSS** about 20 pages or 1 chapter per week	
WEEK 11	⬚ **45m Introduction to Engineering Lessons - Lesson 55** *Present What You've Learned* **→ PREPARE** For one of your exam questions, you will be asked to give an informal presentation to your teacher on something you have learned about this term. This could be how a particular object is made or constructed, or something else you have researched. Use this time to gather your thoughts, think through how you want to do that, and work on it. You can be as creative as you would like.	
WEEK 11	⬚ **60m Introduction to Engineering Labs - Lesson 11** Materials: by student choice & notebook **→ PROJECT**	

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Term 1

Continue working on your project.

WEEK 12

⬚ **45m Introduction to Engineering Lessons - Lesson 56**
Exam Week

➔ EXAMS
Answer question(s) related to course.

WEEK 12

⬚ **45m Introduction to Engineering Lessons - Lesson 57**
Exam Week

➔ EXAMS
Answer question(s) related to course.

WEEK 12

⬚ **45m Introduction to Engineering Lessons - Lesson 58**
Exam Week

➔ EXAMS
Answer question(s) related to course.

WEEK 12

⬚ **45m Introduction to Engineering Lessons - Lesson 59**
Exam Week

➔ EXAMS
Answer question(s) related to course.

WEEK 12

⬚ **45m Introduction to Engineering Lessons - Lesson 60**
Exam Week

➔ EXAMS
Answer question(s) related to course.

WEEK 12

⬚ **60m Introduction to Engineering Labs - Lesson 12**
Exam Week

➔ EXAMS
Answer exam questions from:
Project Work

Science: Introduction to Engineering

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QR

Term 2

WEEK 1

⬚ **45m Introduction to Engineering Lessons - Lesson 61**
How Engineering Works

Materials: The Ways Things Work Now

→ INTRO
This approachable book will help you better understand how everyday things work. The nature of the book is not to be read like a story from cover to cover, but to explore at your own pace. Take as much or as little time as you like with each item.

→ READ, NARRATE, & DISCUSS
from Part 1

• PLAN WEEKLY
⬚ take a nature walk
⬚ science free read

WEEK 1

⬚ **45m Introduction to Engineering Lessons - Lesson 62**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 1

⬚ **45m Introduction to Engineering Lessons - Lesson 63**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 1

⬚ **45m Introduction to Engineering Lessons - Lesson 64**
Engineering Literature

Materials: The Boy Who Harnessed the Wind

→ NOTE
We'll finish this book in week 4, so think about how much you have left and decide if you need to read from it during your free read time.

→ READ, NARRATE, & DISCUSS
about 20 pages or 1 chapter per week

WEEK 1

⬚ **45m Introduction to Engineering Lessons - Lesson 65**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 1

⬚ **60m Introduction to Engineering Labs - Lesson 13**

Materials: by student choice & notebook

→ PROJECT
If there is a design problem in your home or community that you would like to tackle, use this time to research, design, and execute your solution.

If you'd like to sample different areas of engineering, try the mechanical engineering project linked, Choosing the Best Gear Ratio for Maximum Bike Speed. Or you might find your own idea by exploring the Science Buddies website. The filters on the left side of the page allow you to narrow your results. Once you have chosen a project, let your teacher know if you need any materials. Your project might take the whole year or you might finish it in just a few weeks. Whenever you finish your project, select another one! Record your work and your observations in your lab notebook.
∞ Website Link: Choosing the Best Gear Ratio for Maximum Bike Speed
∞ Website Link: Science Buddies

WEEK 2

⬚ **45m Introduction to Engineering Lessons - Lesson 66**
How Engineering Works

Materials: The Ways Things Work Now

→ READ, NARRATE, & DISCUSS

• PLAN WEEKLY
⬚ take a nature walk
⬚ science free read

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QR

Term 2

from Part 1

WEEK 2

⬚ **45m Introduction to Engineering Lessons - Lesson 67**
Tinker Lab

▯ Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 2

⬚ **45m Introduction to Engineering Lessons - Lesson 68**
Tinker Lab

▯ Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 2

⬚ **45m Introduction to Engineering Lessons - Lesson 69**
Engineering Literature

▯ Materials: The Boy Who Harnessed the Wind

→ READ, NARRATE, & DISCUSS
about 20 pages or 1 chapter per week

WEEK 2

⬚ **45m Introduction to Engineering Lessons - Lesson 70**
Flex Day

▯ Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 2

⬚ **60m Introduction to Engineering Labs - Lesson 14**

▯ Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 3

⬚ **45m Introduction to Engineering Lessons - Lesson 71**
How Engineering Works

▯ Materials: The Ways Things Work Now

→ READ, NARRATE, & DISCUSS
from Part 1

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 3

⬚ **45m Introduction to Engineering Lessons - Lesson 72**
Tinker Lab

▯ Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 3

⬚ **45m Introduction to Engineering Lessons - Lesson 73**
Tinker Lab

▯ Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 3

⬚ **45m Introduction to Engineering Lessons - Lesson 74**
Engineering Literature

▯ Materials: The Boy Who Harnessed the Wind

→ READ, NARRATE, & DISCUSS
about 20 pages or 1 chapter per week

Science: Introduction to Engineering

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Term 2

WEEK 3

⬚ **45m Introduction to Engineering Lessons - Lesson 75**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 3

⬚ **60m Introduction to Engineering Labs - Lesson 15**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 4

⬚ **45m Introduction to Engineering Lessons - Lesson 76**
How Engineering Works

Materials: The Ways Things Work Now

→ READ, NARRATE, & DISCUSS
from Part 1 or move on to Part 2, if you are ready.

- **PLAN WEEKLY**
- ⬚ take a nature walk
- ⬚ science free read

WEEK 4

⬚ **45m Introduction to Engineering Lessons - Lesson 77**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 4

⬚ **45m Introduction to Engineering Lessons - Lesson 78**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 4

⬚ **45m Introduction to Engineering Lessons - Lesson 79**
Engineering Literature

Materials: The Boy Who Harnessed the Wind

→ READ, NARRATE, & DISCUSS
about 20 pages or 1 chapter per week

WEEK 4

⬚ **45m Introduction to Engineering Lessons - Lesson 80**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 4

⬚ **60m Introduction to Engineering Labs - Lesson 16**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 5

⬚ **45m Introduction to Engineering Lessons - Lesson 81**
How Engineering Works

Materials: The Ways Things Work Now

→ READ, NARRATE, & DISCUSS
from Part 2

- **PLAN WEEKLY**
- ⬚ take a nature walk
- ⬚ science free read

Science: Introduction to Engineering

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Term 2

WEEK 5

⬚ **45m Introduction to Engineering Lessons - Lesson 82**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 5

⬚ **45m Introduction to Engineering Lessons - Lesson 83**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 5

⬚ **45m Introduction to Engineering Lessons - Lesson 84**
Design Conversations

Materials: website links

→ INTRO
The series of videos (this week and next) titled Nature as Measure is a conversation between Wes Jackson and Wendell Berry regarding problems in agriculture, considering nature as a design model, and choosing a design to solve long-term problems rather than create them.

→ VIEW, NARRATE, & DISCUSS
∞ Video Link: Nature as Measure Part 1 (14:46)
∞ Video Link: Nature as Measure Part 2 (13:58)

WEEK 5

⬚ **45m Introduction to Engineering Lessons - Lesson 85**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 5

⬚ **60m Introduction to Engineering Labs - Lesson 17**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 6

⬚ **45m Introduction to Engineering Lessons - Lesson 86**
How Engineering Works

Materials: The Ways Things Work Now

→ READ, NARRATE, & DISCUSS
from Part 2

• PLAN WEEKLY
⬚ take a nature walk
⬚ science free read

WEEK 6

⬚ **45m Introduction to Engineering Lessons - Lesson 87**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 6

⬚ **45m Introduction to Engineering Lessons - Lesson 88**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

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Term 2

WEEK 6

☐ **45m Introduction to Engineering Lessons - Lesson 89**
Design Conversations

Materials: website links

→ VIEW, NARRATE, & DISCUSS
∞ Video Link: Nature as Measure Part 3 (14:38)
∞ Video Link: Nature as Measure Part 4 (15:47)

WEEK 6

☐ **45m Introduction to Engineering Lessons - Lesson 90**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 6

☐ **60m Introduction to Engineering Labs - Lesson 18**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 7

☐ **45m Introduction to Engineering Lessons - Lesson 91**
How Engineering Works

Materials: The Ways Things Work Now

→ READ, NARRATE, & DISCUSS
from Part 2 or move on to Part 3, if you are ready.

• **PLAN WEEKLY**
☐ take a nature walk
☐ science free read

WEEK 7

☐ **45m Introduction to Engineering Lessons - Lesson 92**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 7

☐ **45m Introduction to Engineering Lessons - Lesson 93**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 7

☐ **45m Introduction to Engineering Lessons - Lesson 94**
Design Conversations

Materials: website link

→ INTRO
Have you thought about nature as a measure in building design? Can you apply the same measure to other applications of design? Let's see what today's speaker has done with this idea.

→ VIEW, NARRATE, & DISCUSS
∞ Video Link: Using Nature's Genius in Architecture (16:55)

WEEK 7

☐ **45m Introduction to Engineering Lessons - Lesson 95**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

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Term 2

WEEK 7

⬚ **60m Introduction to Engineering Labs - Lesson 19**

▯ Materials: by student choice & notebook

→ PROJECT
If there is a design problem in your home or community that you would like to tackle, use this time to research, design, and execute your solution.

If you'd like to sample different areas of engineering, try the electronics engineering project linked, Make Your Own Low-Power AM Radio Transmitter. Or you might find your own idea by exploring the Science Buddies website. The filters on the left side of the page allow you to narrow your results. Once you have chosen a project, let your teacher know if you need any materials. Your project might take the whole year or you might finish it in just a few weeks. Whenever you finish your project, select another one! Record your work and your observations in your lab notebook.
∞ Website Link: Make Your Own Low-Power AM Radio Transmitter
∞ Website Link: Science Buddies

WEEK 8

⬚ **45m Introduction to Engineering Lessons - Lesson 96**
How Engineering Works

▯ Materials: The Ways Things Work Now

→ READ, NARRATE, & DISCUSS
from Part 3

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 8

⬚ **45m Introduction to Engineering Lessons - Lesson 97**
Tinker Lab

▯ Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 8

⬚ **45m Introduction to Engineering Lessons - Lesson 98**
Tinker Lab

▯ Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 8

⬚ **45m Introduction to Engineering Lessons - Lesson 99**
Design Challenge

▯ Materials: library/internet access, possibly a site visit sometime in the next week

→ INTRO
With these design ideas in mind that you have been learning about, spend the last few weeks of this term considering how a design problem might be solved using nature as the inspiration. You are not expected to execute the solution; only to think about worthy design ideas.

→ CONSIDER AND RESEARCH
Water mismanagement is a common problem. Sometimes it is caused by poorly designed landscapes or land use. Sometimes it is caused by inefficient water usage systems. Study a location known to you, such as your home, school, church, grocery store, etc. Learn how water is used at that location, what happens to water when it rains, etc. Learn how water is used in local habitats. Are there large quantities that nature must channel? Is there a scarcity that nature must conserve? How do plants and animals access water? Visit your location and/or a local nature preserve, if needed. You will have another week to complete your research before continuing the project.

WEEK 8

⬚ **45m Introduction to Engineering Lessons - Lesson 100**
Flex Day

▯ Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

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Term 2

WEEK 8

☐ **60m Introduction to Engineering Labs - Lesson 20**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 9

☐ **45m Introduction to Engineering Lessons - Lesson 101**
How Engineering Works

Materials: The Ways Things Work Now

→ READ, NARRATE, & DISCUSS
from Part 3

• **PLAN WEEKLY**
☐ take a nature walk
☐ science free read

WEEK 9

☐ **45m Introduction to Engineering Lessons - Lesson 102**
Tinker Lab

Materials: by student choice

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 9

☐ **45m Introduction to Engineering Lessons - Lesson 103**
Tinker Lab

Materials: by student choice

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 9

☐ **45m Introduction to Engineering Lessons - Lesson 104**
Design Challenge

Materials: library/internet access, possibly a site visit

→ RESEARCH
Continue researching your design challenge this week. Visit your location and/or a local nature preserve, if needed.

WEEK 9

☐ **45m Introduction to Engineering Lessons - Lesson 105**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 9

☐ **60m Introduction to Engineering Labs - Lesson 21**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 10

☐ **45m Introduction to Engineering Lessons - Lesson 106**
How Engineering Works

Materials: The Ways Things Work Now

→ READ, NARRATE, & DISCUSS
from Part 3

• **PLAN WEEKLY**
☐ take a nature walk
☐ science free read

WEEK 10

☐ **45m Introduction to Engineering Lessons - Lesson 107**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

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Term 2

WEEK 10

⬚ 45m Introduction to Engineering Lessons - Lesson 108

Tinker Lab

Materials: by student choice & notebook

➔ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 10

⬚ 45m Introduction to Engineering Lessons - Lesson 109

Design Challenge

Materials: library/internet access & notebook

➔ BRAINSTORM
Make a list of ideas that could help address the water management issues at your location. Intentionally consider how water management in the local habitat might point to solutions at your location. When you are finished brainstorming (today or tomorrow), begin working out possible design solutions.

WEEK 10

⬚ 45m Introduction to Engineering Lessons - Lesson 110

Design Challenge

Materials: library/internet access & notebook

➔ CHOOSE A SOLUTION
Choose one nature-inspired solution to the water management problem. You want to understand:
1. the problem,
2. nature's solution, and
3. how nature's solution could improve your location.

You will have time to finish this next week before preparing an informal presentation for your teacher.

WEEK 10

⬚ 60m Introduction to Engineering Labs - Lesson 22

Materials: by student choice & notebook

➔ PROJECT
Continue working on your project.

WEEK 11

⬚ 45m Introduction to Engineering Lessons - Lesson 111

How Engineering Works

Materials: The Ways Things Work Now

➔ READ, NARRATE, & DISCUSS
from Part 3

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 11

⬚ 45m Introduction to Engineering Lessons - Lesson 112

Tinker Lab

Materials: by student choice & notebook

➔ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 11

⬚ 45m Introduction to Engineering Lessons - Lesson 113

Tinker Lab

Materials: by student choice & notebook

➔ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 11

⬚ 45m Introduction to Engineering Lessons - Lesson 114

Present What You've Learned

➔ PREPARE
For one of your exam questions, you will be asked to give an informal presentation to your teacher on your solution to the design challenge. If you are still working on your idea, finish it today. When you are finished, create a poster, Canva graphic, or other presentation method to show:
1. the problem,
2. nature's solution, and
3. how nature's solution could improve your location.

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Term 2

WEEK 11

⬚ **45m Introduction to Engineering Lessons - Lesson 115**
Present What You've Learned

→ PREPARE
Complete your project. Create a poster, Canva graphic, or use another presentation method to show what you've learned.

WEEK 11

⬚ **60m Introduction to Engineering Labs - Lesson 23**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 12

⬚ **45m Introduction to Engineering Lessons - Lesson 116**
Exam Week

→ EXAMS
Answer question(s) related to course.

WEEK 12

⬚ **45m Introduction to Engineering Lessons - Lesson 117**
Exam Week

→ EXAMS
Answer question(s) related to course.

WEEK 12

⬚ **45m Introduction to Engineering Lessons - Lesson 118**
Exam Week

→ EXAMS
Answer question(s) related to course.

WEEK 12

⬚ **45m Introduction to Engineering Lessons - Lesson 119**
Exam Week

→ EXAMS
Answer question(s) related to course.

WEEK 12

⬚ **45m Introduction to Engineering Lessons - Lesson 120**
Exam Week

→ EXAMS
Answer question(s) related to course.

WEEK 12

⬚ **60m Introduction to Engineering Labs - Lesson 24**
Exam Week

→ EXAMS
Answer exam questions from:
Project Work

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Term 3

WEEK 1

⬚ **45m Introduction to Engineering Lessons - Lesson 121**
How Engineering Works

Materials: The Ways Things Work Now

→ READ, NARRATE, & DISCUSS
from Part 4

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 1

⬚ **45m Introduction to Engineering Lessons - Lesson 122**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 1

⬚ **45m Introduction to Engineering Lessons - Lesson 123**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 1

⬚ **45m Introduction to Engineering Lessons - Lesson 124**
The Ethics of Design

Materials: By Design

→ INTRO
This is a challenging book written by a Professor of Theology for engineering students at the University of Dayton. This is important for two reasons: 1. You might not yet understand some of the math and science that he references. 2. You might not agree with his particular theological beliefs. Neither math and science nor his religion is the point, however. The point is that we think about design within the context of human living. This is important to think about and is often overlooked in higher training, so it's a good time to get an introduction now, but don't get bogged down. Read what you can this term (the schedule is only a suggestion); if you don't end up finishing it now, you can come back to it later, knowing that these important ideas are waiting for you.

→ READ, NARRATE, & DISCUSS
p.1-12

WEEK 1

⬚ **45m Introduction to Engineering Lessons - Lesson 125**
The Ethics of Design

Materials: By Design

→ READ, NARRATE, & DISCUSS
p.12-25. Explore the Discussion Questions and Notes only as desired.

WEEK 1

⬚ **60m Introduction to Engineering Labs - Lesson 25**

Materials: by student choice & notebook

→ PROJECT
If there is a design problem in your home or community that you would like to tackle, use this time to research, design, and execute your solution.

Or you might choose between the linked materials (Living Loom), electronic engineering projects(Obstacle-Detecting Cane with Arduino), or another project from the Science Buddies website. Once you have chosen a project, let your teacher know if you need any materials.
∞ Website Link: Living Loom
∞ Website Link: Obstacle-Detecting Cane with Arduino
∞ Website Link: Science Buddies

WEEK 2

⬚ **45m Introduction to Engineering Lessons - Lesson 126**
How Engineering Works

Materials: The Ways Things Work Now

→ READ, NARRATE, & DISCUSS
from Part 4

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

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Term 3

WEEK 2

⬚ **45m Introduction to Engineering Lessons - Lesson 127**
Tinker Lab

▯ Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 2

⬚ **45m Introduction to Engineering Lessons - Lesson 128**
Tinker Lab

▯ Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 2

⬚ **45m Introduction to Engineering Lessons - Lesson 129**
The Ethics of Design

▯ Materials: By Design

→ NOTE
There is no Fig.2.1 in Ch.2. Refer to Fig.1.7 instead.

→ READ, NARRATE, & DISCUSS
p.31-38

WEEK 2

⬚ **45m Introduction to Engineering Lessons - Lesson 130**
Flex Day

▯ Materials: Selected engineering literature, tinker/project supplies, or internet access

→ NOTE
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 2

⬚ **60m Introduction to Engineering Labs - Lesson 26**

▯ Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 3

⬚ **45m Introduction to Engineering Lessons - Lesson 131**
How Engineering Works

▯ Materials: The Ways Things Work Now

→ READ, NARRATE, & DISCUSS
from Part 4

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 3

⬚ **45m Introduction to Engineering Lessons - Lesson 132**
Tinker Lab

▯ Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 3

⬚ **45m Introduction to Engineering Lessons - Lesson 133**
Tinker Lab

▯ Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 3

⬚ **45m Introduction to Engineering Lessons - Lesson 134**
The Ethics of Design

▯ Materials: By Design

→ NOTE
The reference to Fig.2.2 toward the end of p.48 should be Fig.2.5.

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Term 3

→ **READ, NARRATE, & DISCUSS**
p.38-49 Explore the Discussion Questions and Notes only as desired.

WEEK 3

⬚ **45m Introduction to Engineering Lessons - Lesson 135**
The Ethics of Design

Materials: By Design

→ **READ, NARRATE, & DISCUSS**
p.53-65

WEEK 3

⬚ **60m Introduction to Engineering Labs - Lesson 27**

Materials: by student choice & notebook

→ **PROJECT**
Continue working on your project.

WEEK 4

⬚ **45m Introduction to Engineering Lessons - Lesson 136**
How Engineering Works

Materials: The Ways Things Work Now

→ **READ, NARRATE, & DISCUSS**
from Part 4

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 4

⬚ **45m Introduction to Engineering Lessons - Lesson 137**
Tinker Lab

Materials: by student choice & notebook

→ **TINKER**
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 4

⬚ **45m Introduction to Engineering Lessons - Lesson 138**
Tinker Lab

Materials: by student choice & notebook

→ **TINKER**
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 4

⬚ **45m Introduction to Engineering Lessons - Lesson 139**
The Ethics of Design

Materials: By Design

→ **READ, NARRATE, & DISCUSS**
p.65-77 Explore the Discussion Questions and Notes only as desired.

WEEK 4

⬚ **45m Introduction to Engineering Lessons - Lesson 140**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ **NOTE**
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 4

⬚ **60m Introduction to Engineering Labs - Lesson 28**

Materials: by student choice & notebook

→ **PROJECT**
Continue working on your project.

WEEK 5

⬚ **45m Introduction to Engineering Lessons - Lesson 141**
How Engineering Works

Materials: The Ways Things Work Now

→ **READ, NARRATE, & DISCUSS**

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

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Term 3

from Part 4 or move on to Part 5, if you are ready.

WEEK 5

⬚ **45m Introduction to Engineering Lessons - Lesson 142**
Tinker Lab

▯ Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 5

⬚ **45m Introduction to Engineering Lessons - Lesson 143**
Tinker Lab

▯ Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 5

⬚ **45m Introduction to Engineering Lessons - Lesson 144**
The Ethics of Design

▯ Materials: By Design

→ NOTE
We are skipping Ch.4. You may read it during your free read time, if desired.

→ READ, NARRATE, & DISCUSS
p.101-109

WEEK 5

⬚ **45m Introduction to Engineering Lessons - Lesson 145**
The Ethics of Design

▯ Materials: By Design

→ READ, NARRATE, & DISCUSS
p.109-118 Explore the Discussion Questions and Notes only as desired.

WEEK 5

⬚ **60m Introduction to Engineering Labs - Lesson 29**

▯ Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 6

⬚ **45m Introduction to Engineering Lessons - Lesson 146**
How Engineering Works

▯ Materials: The Ways Things Work Now

→ READ, NARRATE, & DISCUSS
from Part 5

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 6

⬚ **45m Introduction to Engineering Lessons - Lesson 147**
Tinker Lab

▯ Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 6

⬚ **45m Introduction to Engineering Lessons - Lesson 148**
Tinker Lab

▯ Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 6

⬚ **45m Introduction to Engineering Lessons - Lesson 149**
The Ethics of Design

▯ Materials: By Design

→ READ, NARRATE, & DISCUSS
p.121-131

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Term 3

WEEK 6

⬚ **45m Introduction to Engineering Lessons - Lesson 150**

Flex Day

▯ Materials: Selected engineering literature, tinker/project supplies, or internet access

➜ NOTE

Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 6

⬚ **60m Introduction to Engineering Labs - Lesson 30**

▯ Materials: by student choice & notebook

➜ PROJECT

If there is a design problem in your home or community that you would like to tackle, use this time to research, design, and execute your solution.

Or you might choose between the linked chemical formulation (Design Your Own Sports Drink), electronics engineering (Build a Magnetic Pet Door with Arduino) projects, or another project from the Science Buddies website. Note that the chemistry project only becomes an engineering project by extending it into the design phase! Once you have chosen a project, let your teacher know if you need any materials.

∞ Website Link: Design Your Own Sports Drink
∞ Website Link: Build a Magnetic Pet Door with Arduino
∞ Website Link: Science Buddies

WEEK 7

⬚ **45m Introduction to Engineering Lessons - Lesson 151**

How Engineering Works

▯ Materials: The Ways Things Work Now

➜ READ, NARRATE, & DISCUSS

from Part 5

• PLAN WEEKLY

⬚ take a nature walk
⬚ science free read

WEEK 7

⬚ **45m Introduction to Engineering Lessons - Lesson 152**

Tinker Lab

▯ Materials: by student choice & notebook

➜ TINKER

Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 7

⬚ **45m Introduction to Engineering Lessons - Lesson 153**

Tinker Lab

▯ Materials: by student choice & notebook

➜ TINKER

Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 7

⬚ **45m Introduction to Engineering Lessons - Lesson 154**

The Ethics of Design

▯ Materials: By Design

➜ READ, NARRATE, & DISCUSS

p.131-142 Explore the Discussion Questions and Notes only as desired.

WEEK 7

⬚ **45m Introduction to Engineering Lessons - Lesson 155**

The Ethics of Design

▯ Materials: By Design

➜ NOTE

There is an important discussion in this chapter regarding habits, virtue, and character. The author includes a phrase regarding "extenuating circumstances." It is not that extenuating circumstances are an excuse for bad behavior, but rather that sometimes behavior might appear to reflect bad character when it really reflects an injury, a developmental difference, or that the person comes from a different culture with different norms. As you read this chapter, consider how your character might be (or maybe already has been) misconstrued, if you are someone with such extenuating circumstances - or if someday you become someone with such circumstances. How can you continue to build good character when your actions might be misconstrued? If someone in your community displays surprising or apparently inconsistent character, how might you respond generously and compassionately? The author might have some ideas to help, so keep reading and see what you think.

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Term 3

→ **READ, NARRATE, & DISCUSS**
p.147-162

WEEK 7

⬚ **60m Introduction to Engineering Labs - Lesson 31**

Materials: by student choice & notebook

→ **PROJECT**
Continue working on your project.

WEEK 8

⬚ **45m Introduction to Engineering Lessons - Lesson 156**
How Engineering Works

Materials: The Ways Things Work Now

→ **READ, NARRATE, & DISCUSS**
from Part 5

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 8

⬚ **45m Introduction to Engineering Lessons - Lesson 157**
Tinker Lab

Materials: by student choice & notebook

→ **TINKER**
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 8

⬚ **45m Introduction to Engineering Lessons - Lesson 158**
Tinker Lab

Materials: by student choice & notebook

→ **TINKER**
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 8

⬚ **45m Introduction to Engineering Lessons - Lesson 159**
The Ethics of Design

Materials: By Design

→ **READ, NARRATE, & DISCUSS**
p.163-176 Explore the Discussion Questions and Notes only as desired.

WEEK 8

⬚ **45m Introduction to Engineering Lessons - Lesson 160**
Flex Day

Materials: Selected engineering literature, tinker/project supplies, or internet access

→ **NOTE**
Use this flex day to finish the week's reading or continue any hands-on activities. Or, you might read and narrate a current event from Science News Explores (in your Quick Links) or research an interesting engineering question that came to mind during a nature walk or drive around town, such as 'How do frogs jump so far?' or 'Why does that tall building look like it's tilting?'

WEEK 8

⬚ **60m Introduction to Engineering Labs - Lesson 32**

Materials: by student choice & notebook

→ **PROJECT**
Continue working on your project.

WEEK 9

⬚ **45m Introduction to Engineering Lessons - Lesson 161**
How Engineering Works

Materials: The Ways Things Work Now

→ **READ, NARRATE, & DISCUSS**
from Part 5

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 9

⬚ **45m Introduction to Engineering Lessons - Lesson 162**
Tinker Lab

Materials: by student choice & notebook

→ **TINKER**

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Term 3

Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 9

☐ 45m Introduction to Engineering Lessons - Lesson 163
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 9

☐ 45m Introduction to Engineering Lessons - Lesson 164
The Ethics of Design

Materials: By Design

→ NOTE
The author discusses the difference between humans and other animals in Ch.8. Consider whether you agree. Do you have anything to add, question, or clarify?

→ READ, NARRATE, & DISCUSS
p.182-193

WEEK 9

☐ 45m Introduction to Engineering Lessons - Lesson 165
The Ethics of Design

Materials: By Design

→ READ, NARRATE, & DISCUSS
p.193-201 Explore the Discussion Questions and Notes only as desired.

WEEK 9

☐ 60m Introduction to Engineering Labs - Lesson 33

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 10

☐ 45m Introduction to Engineering Lessons - Lesson 166
How Engineering Works

Materials: The Ways Things Work Now

→ READ, NARRATE, & DISCUSS
from Part 5

- **PLAN WEEKLY**
  - ☐ take a nature walk
  - ☐ science free read

WEEK 10

☐ 45m Introduction to Engineering Lessons - Lesson 167
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 10

☐ 45m Introduction to Engineering Lessons - Lesson 168
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 10

☐ 45m Introduction to Engineering Lessons - Lesson 169
The Ethics of Design

Materials: By Design

→ NOTE
We are skipping Ch.9. You may read it during your free read time, if desired.

→ READ, NARRATE, & DISCUSS
p.248-259

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Term 3

WEEK 10

⬚ **45m Introduction to Engineering Lessons - Lesson 170**
Present What You've Learned

→ PREPARE
For one of your exam questions, you will be asked to give an informal presentation to your teacher on something you have learned about this term. This could be how a particular object is made or constructed, or something else you have researched. Use this time to gather your thoughts and to think through how you want to do that. You can be as creative as you would like.

WEEK 10

⬚ **60m Introduction to Engineering Labs - Lesson 34**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 11

⬚ **45m Introduction to Engineering Lessons - Lesson 171**
How Engineering Works

Materials: The Ways Things Work Now

→ READ, NARRATE, & DISCUSS
from Part 5

• **PLAN WEEKLY**
⬚ take a nature walk
⬚ science free read

WEEK 11

⬚ **45m Introduction to Engineering Lessons - Lesson 172**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 11

⬚ **45m Introduction to Engineering Lessons - Lesson 173**
Tinker Lab

Materials: by student choice & notebook

→ TINKER
Take apart, figure out, and attempt to put back together (or improve). Record your observations.

WEEK 11

⬚ **45m Introduction to Engineering Lessons - Lesson 174**
The Ethics of Design

Materials: By Design

→ READ, NARRATE, & DISCUSS
p.259-269 Explore the Discussion Questions and Notes only as desired.

WEEK 11

⬚ **45m Introduction to Engineering Lessons - Lesson 175**
Present What You've Learned

→ PREPARE
For one of your exam questions, you will be asked to give an informal presentation to your teacher on something you have learned about this term. This could be how a particular object is made or constructed, or something else you have researched. Use this time to gather your thoughts and to think through how you want to do that. You can be as creative as you would like.

WEEK 11

⬚ **60m Introduction to Engineering Labs - Lesson 35**

Materials: by student choice & notebook

→ PROJECT
Continue working on your project.

WEEK 12

⬚ **45m Introduction to Engineering Lessons - Lesson 176**
Exam Week

→ EXAMS
Answer question(s) related to course.

WEEK 12

⬚ **45m Introduction to Engineering Lessons - Lesson 177**
Exam Week

→ EXAMS
Answer question(s) related to course.

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Term 3

WEEK 12

☐ **45m Introduction to Engineering Lessons - Lesson 178**
Exam Week

→ EXAMS
Answer question(s) related to course.

WEEK 12

☐ **45m Introduction to Engineering Lessons - Lesson 179**
Exam Week

→ EXAMS
Answer question(s) related to course.

WEEK 12

☐ **45m Introduction to Engineering Lessons - Lesson 180**
Exam Week

→ EXAMS
Answer question(s) related to course.

WEEK 12

☐ **60m Introduction to Engineering Labs - Lesson 36**
Exam Week

→ EXAMS
Answer exam questions from:
Project Work

Science: Introduction to Engineering

Examination

Term 1

Introduction to Engineering Lessons

· Tell about an engineering story from The Things We Make.

· The Boy Who Harnessed the Wind: Describe the setting of William's story, including the circumstances that influenced him.

· The Boy Who Harnessed the Wind: Describe William's character, giving examples from the story to support your observations.

· Tell about a nature observation or current event to stood out to you this term.

· Give an informal presentation to your teacher on something you have learned about this term, such as how a particular object is made or constructed, or something else you have researched.

Term 2

Introduction to Engineering Lessons

· Tell about an object you studied in The Way Things Work.

· The Boy Who Harnessed the Wind: Explain how William learned to be a scientist. OR Describe how science and religion both influenced William's life.

· What do we mean by 'Nature as Measure' (i.e. how can nature influence design)? Give at least one example.

· Tell about an object you tinkered with this term. How does it work? What was challenging about the experience? Is there something more you would like to learn about this object?

· Share your solution to the water management design challenge, explaining as needed.

Term 3

Introduction to Engineering Lessons

· Tell about an object you studied in The Way Things Work.

· By Design: How does engineering change the engineer? OR Is engineering a Christian vocation? Why or why not?

· Tell about a nature observation or current event that stood out to you this term.

· Tell about an object you tinkered with this term. How does it work? What was challenging about the experience? Is there something more you would like to learn about this object?

· Give an informal presentation to your teacher on something you have learned about this term, such as how a particular object is made or constructed, or something else you have researched.